

Question:

An 18-year-old college student presents to the ED straight from chemistry lab where he ingested an unknown compound. He complains of a headache, and is flushed, tachypneic and tachycardic. Suspecting cyanide poisoning, you administer amyl nitrite which causes which of the following? Answer Choices: A. Oxidation of ferrous iron in hemoglobin to ferric iron B. A decrease in serum methemoglobin levels C. Formation of thiocyanate D. Increase in intracellular NADH/NAD+ ratio

Reasoning + Answer:

<think> The administration of amyl nitrite is often used to treat cyanide poisoning as it helps to increase the production of thiocyanate, which helps in the excretion of cyanide from the body and reduces its toxic effects on the body's cells by decreasing its binding to the cytochrome aa3 in the mitochondria of cells, which is part of the electron transport chain in cellular respiration. </think> <answer> C . Formation of thiocyanate </answer>

